Hongyan Ji

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Education

The University of Iowa, Iowa City, IA

- Ph.D. in Computer Science 2021–present
  Expected completion: May 2027. Advisor: Sriram V. Pemmaraju.
- Graduate Certificate in College Teaching 2025
- B.S. in Computer Science and Master of Computer Science (M.C.S.) 2017–2021

Research Interests

Algorithms across computational models, especially distributed and parallel graph algorithms; approximation and randomization; communication, memory, and energy efficiency.

Publications

Conference papers

- **Approximating Minimum Dominating Set with Few Awake Rounds.** Hongyan Ji, Shreyas Pai, and Sriram V. Pemmaraju. 40th International Symposium on Distributed Computing (DISC), LIPIcs 397, pp. 53:1–53:21, 2026. Invited to submit the full version to the DISC 2026 special issue of *Distributed Computing*.
- **Fast Deterministic Massively Parallel Ruling Sets Algorithms.** Hongyan Ji, Kishore Kothapalli, Sriram V. Pemmaraju, and Ajitanshu Singh. 26th International Conference on Distributed Computing and Networking (ICDCN), 2025.
- **Faster Set Cover in the MPC Model.** Hongyan Ji, Shreyas Pai, Sriram V. Pemmaraju, and Joshua Sobel. 26th International Conference on Distributed Computing and Networking (ICDCN), 2025. **Best Paper Award, Distributed Computing Track.**
- **Towards Singular Optimality in the Presence of Local Initial Knowledge.** Hongyan Ji and Sriram V. Pemmaraju. 31st International Colloquium on Structural Information and Communication Complexity (SIROCCO), pp. 300–317, 2024.

Journal articles

- **Faster Set Cover in the MPC Model.** Hongyan Ji, Shreyas Pai, Sriram V. Pemmaraju, and Joshua Sobel. *Theoretical Computer Science*, vol. 1086, Article 116238, 2026. Journal version of the ICDCN 2025 paper above.
- **Fast Deterministic Massively Parallel Ruling Sets Algorithms.** Hongyan Ji, Kishore Kothapalli, Sriram V. Pemmaraju, and Ajitanshu Singh. *Theoretical Computer Science*, accepted. Journal version of the ICDCN 2025 paper above.

Honors and Funding

- University of Iowa CS Department Travel Grant ($600), DISC 2026
- University of Iowa Graduate & Professional Student Government (GPSG) Travel Grant ($300) 2026
- **Best Paper Award**, Distributed Computing Track, ICDCN 2025
- ACM PODC Student Travel Grant ($150) 2025
- University of Iowa CS Department Travel Grant ($600), SIROCCO 2024
- NSF I-Corps Workshop, UI Innovators: workshop funding ($1,000) 2023

Teaching Experience

The University of Iowa

- **Sole Instructor**, CS:3210 Programming Languages and Tools: Go — Spring 2026
  Designed and taught a Go course for 18 students; developed the syllabus, lectures, assignments, and assessments around a staged MapReduce project. Used written reflections, handwritten quizzes, and optional oral reassessment to assess students' reasoning and understanding.
- **Guest Instructor**, CS:4330 Theory of Computation — Fall 2024
  Designed and delivered three lectures on context-free grammars through the Practicum in College Teaching.
- **Teaching Assistant**, CS:5620 Distributed Systems and Algorithms — Fall 2024
  Graded assignments and held office hours. Designed and delivered two lectures on network decomposition through the Practicum in College Teaching.
- **Discussion Section Instructor**, CS:2210 Discrete Structures — Spring 2023
  Led two discussion sections, graded homework, and held office hours.
- **Grader**, CS:3330 Algorithms (online section) — Spring 2022
  Graded assignments and held office hours.
- **Grader**, CS:4640 Computer Security — Spring 2021
  Graded assignments and held office hours.

Research Experience

- **Research Assistant**, The University of Iowa. Advisor: Sriram V. Pemmaraju.
  Fall 2021; Fall 2022; Fall 2023; Spring 2024; Spring 2025; Fall 2025; Fall 2026.

Service and Outreach

Professional service

- Conference reviewer: DISC 2026; SOSA 2026; PODC 2025.

Departmental service, The University of Iowa

- Volunteer, CS faculty interviews — Spring 2023 and 2024
- Organizer, Ph.D. Colloquium — Fall 2022
- Board Member, Computer Science Graduate Student Board — Spring 2022

Outreach and mentoring, The University of Iowa

- Panelist, graduate school panel for REU students — Summer 2026
- Mentor, ExploreU STEM Summer Camp — Summers 2021–2023
  Supported mathematical and computational activities for middle school girls.
- Assistant, International Programs – Life in Iowa — 2018–2019

Talks and Presentations

- 40th International Symposium on Distributed Computing (DISC) — Nov 2026 (scheduled)
  Approximating Minimum Dominating Set with Few Awake Rounds. Rome, Italy.
- Algorithms Reading Group, The University of Iowa — Sep 2025
  Efficient Algorithms for Constructing Very Sparse Spanners and Emulators.
- 10th Iowa Computer Science Graduate Research Symposium — Fall 2024
  The University of Iowa.
- 31st International Colloquium on Structural Information and Communication Complexity (SIROCCO) — May 2024
  Towards Singular Optimality in the Presence of Local Initial Knowledge. Vietri sul Mare, Italy.
- Algorithms Reading Group, The University of Iowa — Oct 2023
  Learning-Augmented Algorithms for Online Steiner Tree.